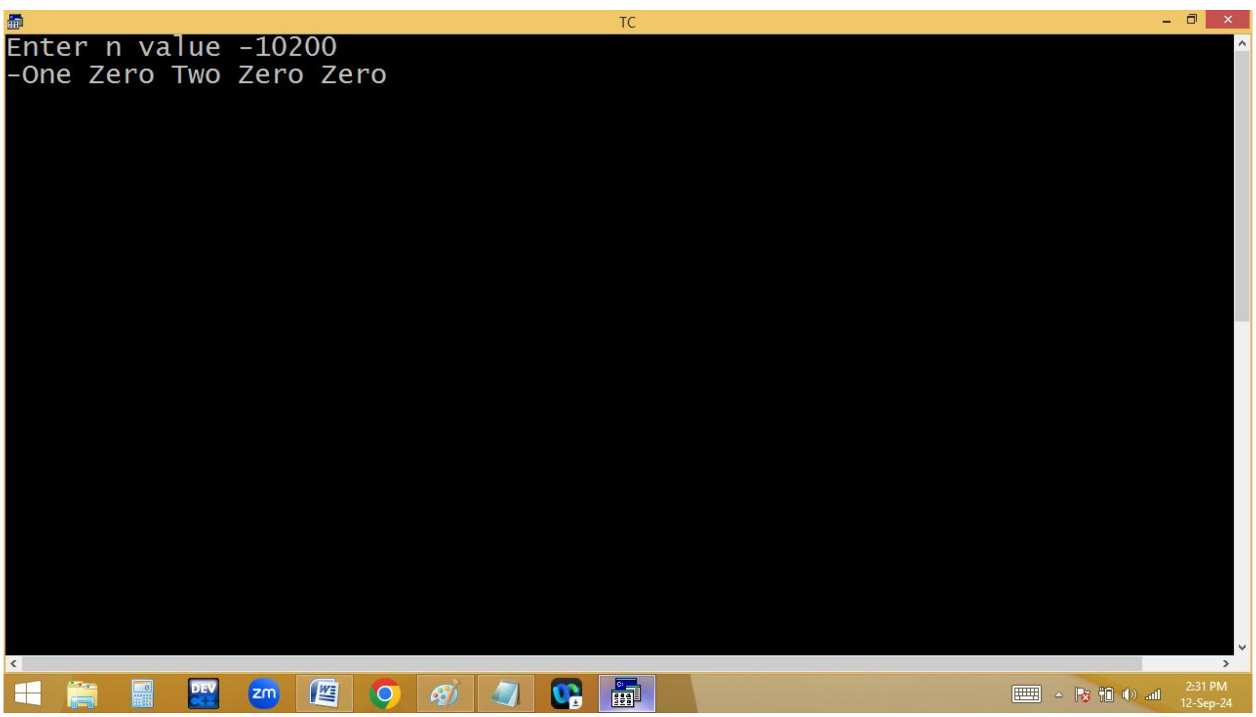
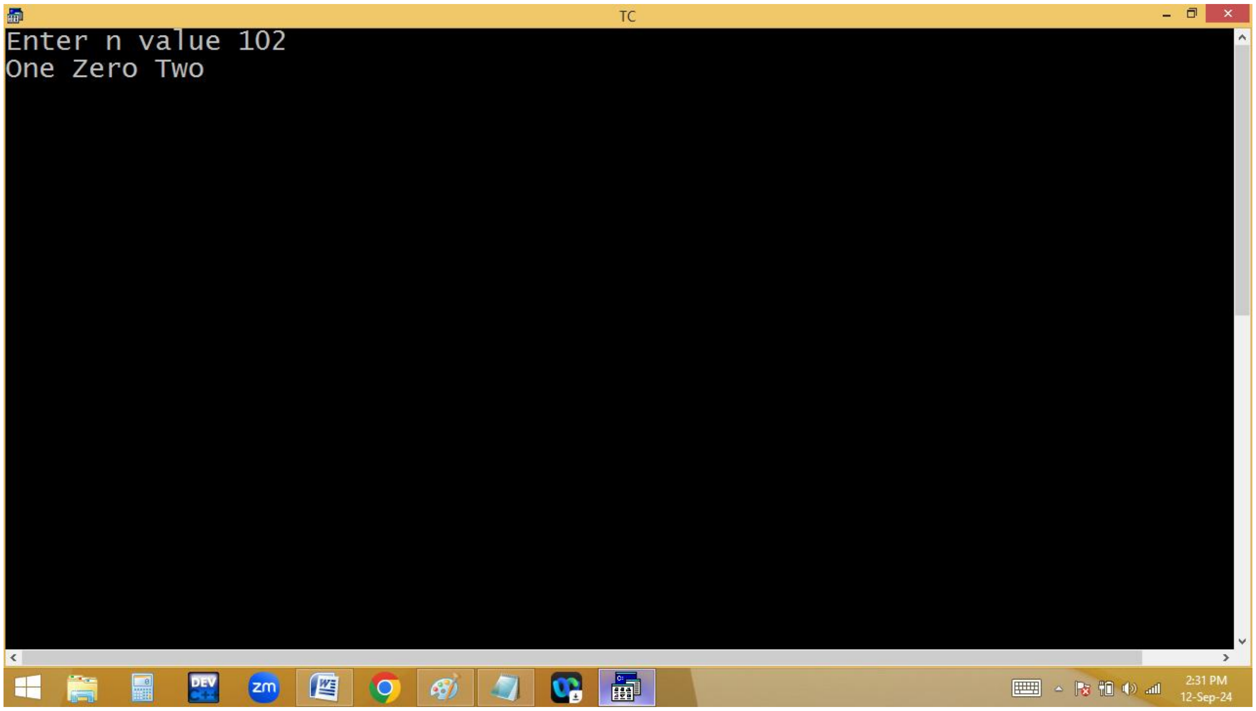
No to text conversion:

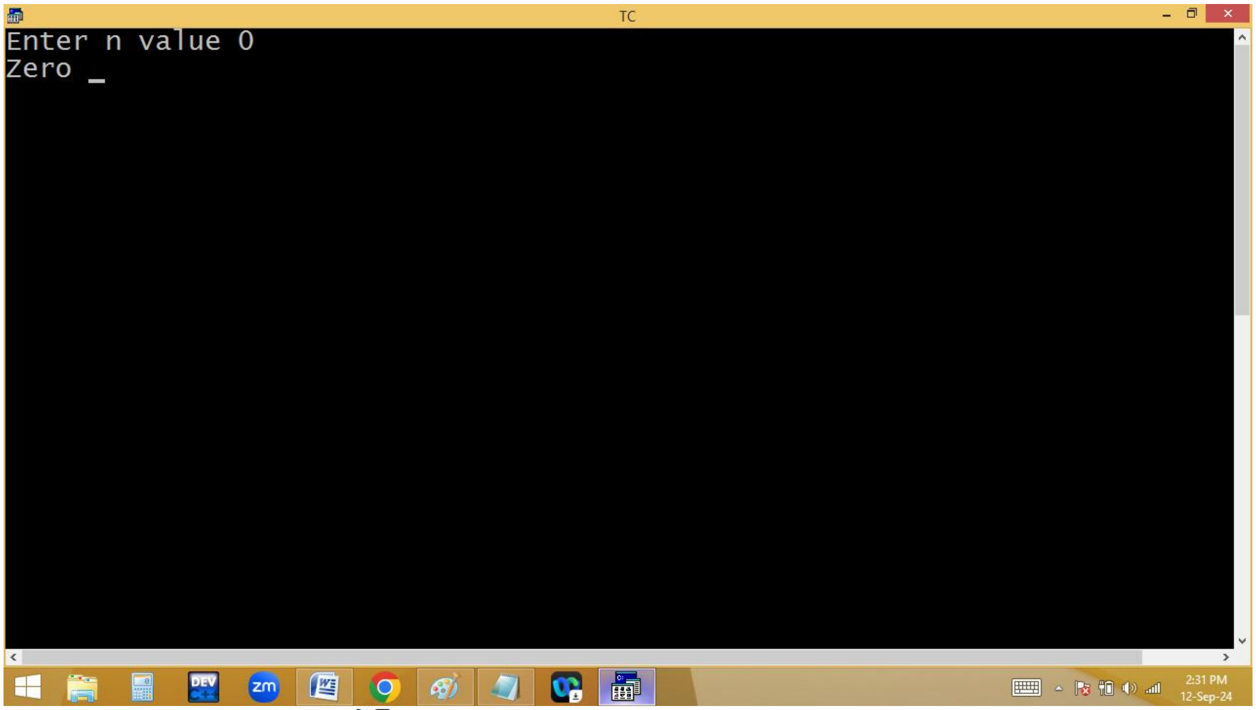
102 → One Zero Two

```
#include<stdio.h>

#include<conio.h>

void main()

{

long n,rev=0,m; int r;

clrscr();

printf("Enter n value "); scanf("%ld",&n);

if(n<0)printf("-",n=-n);  m=n;

while(n!=0){r=n%10; rev=rev*10+r; n=n/10;} /* rev no */

do

{

switch(rev%10)

{

case 0: printf("Zero");break;

case 1: printf("One");break;

case 2: printf("Two");break;

case 3: printf("Three");break;

case 4: printf("Four");break;
```

```
case 5: printf("Five");break;

case 6: printf("Six");break;

case 7: printf("Seven");break;

case 8: printf("Eight");break;

case 9: printf("Nine");

}

printf(" "); rev=rev/10;

}while(rev!=0);

while(m%10==0&&m!=0)printf("Zero ",m=m/10);

getch();

}
```




TC
Enter n value 102
One Zero Two
2:31 PM
12-Sep-24




```
do
{
switch(rev%10)
{
case 0: p(zero);b;
case 1: p(one);b;
case 2: p(two);b;
case 9: p(nine);
}
rev=rev/10; p(" ");
}while(rev!=0);
```

n
102

rev
201 % 10 = 1 One
20 % 10 = 0 Zero
2 % 10 = 2 Two
0

n = m
0

m
100 % 10 = 0

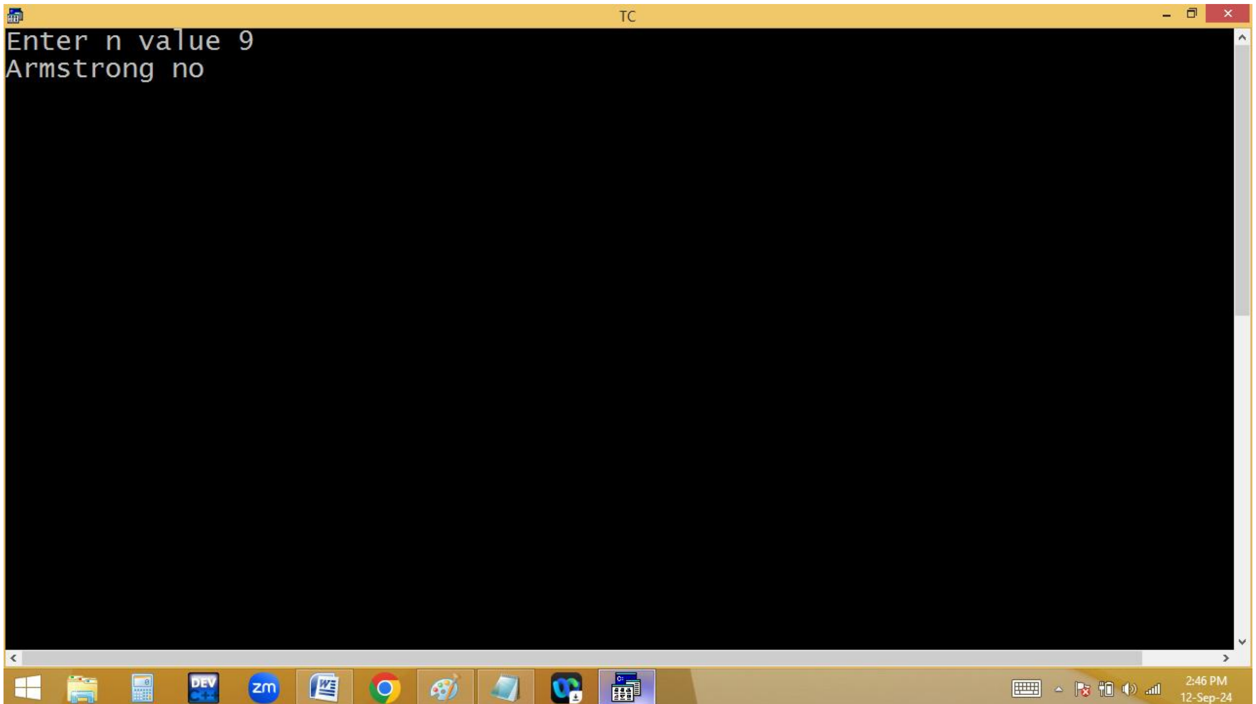
```
while(m%10==0 && m!=0 ) p("Zero ", m=m/10);
```

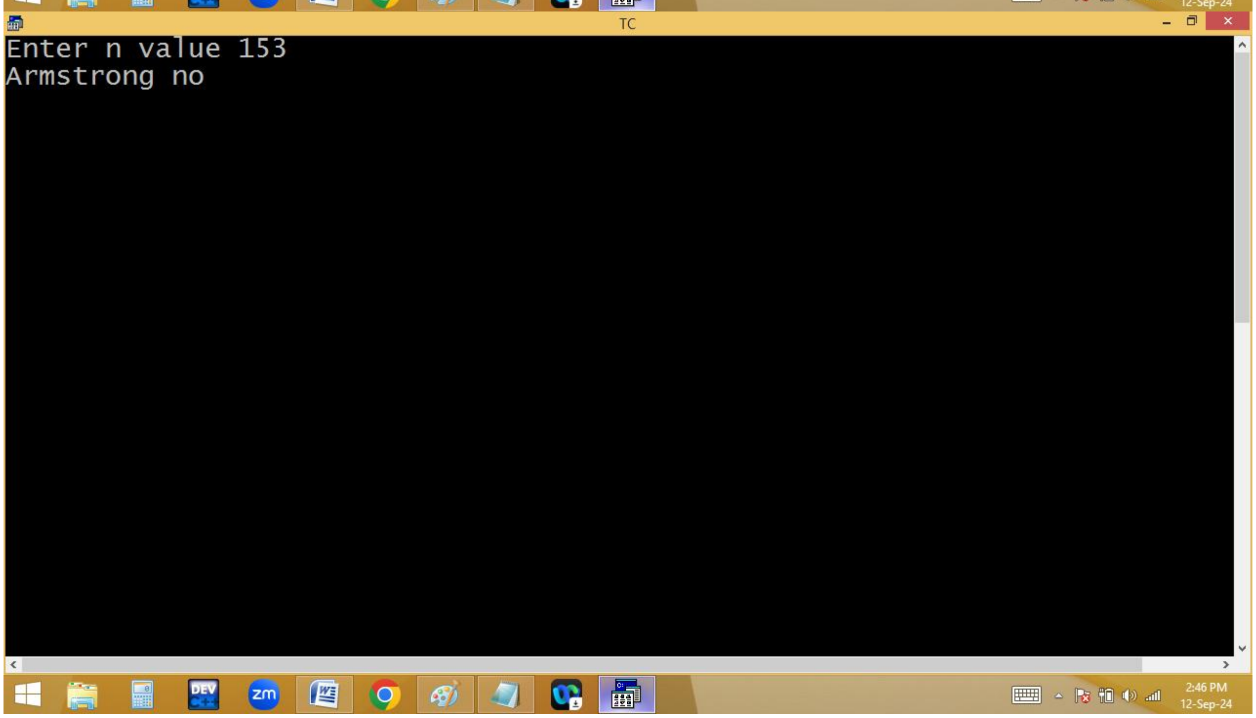
One Zero Zero

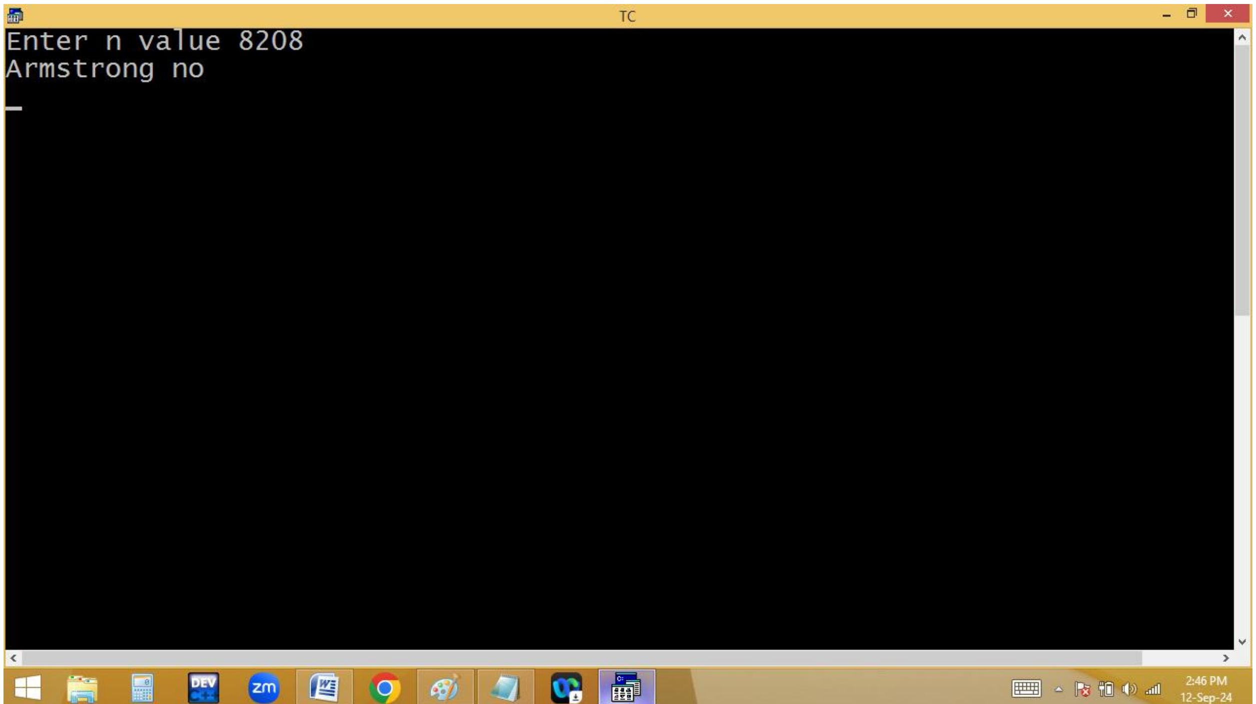
Finding Armstrong no:

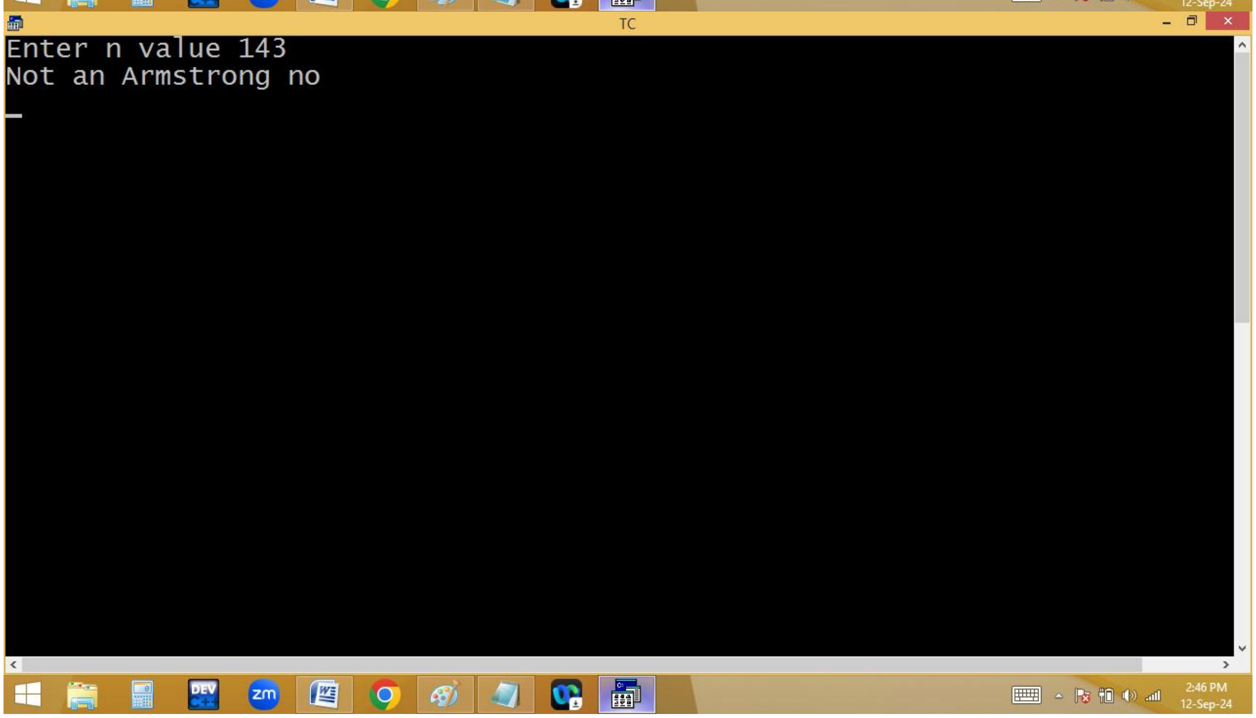
```
File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 14    Col 2    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n,m,r,c=0,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n); m=n;
while(m!=0){c++; m=m/10;} /*counting no of digits */
m=n;
while(m!=0){r=m%10;s+=pow(r,c);m=m/10;}
puts(n==s?"Armstrong no":"Not an Armstrong no");
getch();
}
```

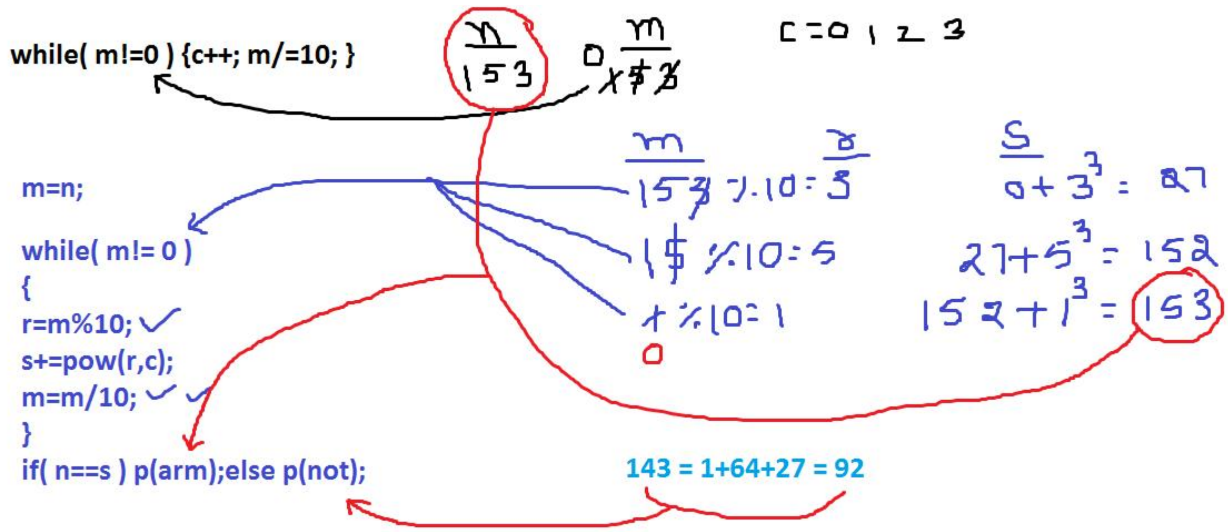
```
Enter n value 1
Armstrong no
```


TC
Enter n value 9
Armstrong no
2:46 PM
12-Sep-24


TC
Enter n value 153
Armstrong no
2:46 PM
12-Sep-24


TC
Enter n value 8208
Armstrong no
2:46 PM
12-Sep-24


TC
Enter n value 143
Not an Armstrong no
2:46 PM
12-Sep-24



for loop:

It is an entry control loop.

for is a keyword.

It is also used to repeat a program several times based on a condition.

When compared with while and do while, for loop is looking to be smart. In for it is compulsory to maintain two semicolons.

For works without condition also and default condition is always 1 i.e. true.

Generally for loop is having 3 expressions.

1. **Initialization**
2. **Test condition / expression**
3. **Increment/decrement / updation**

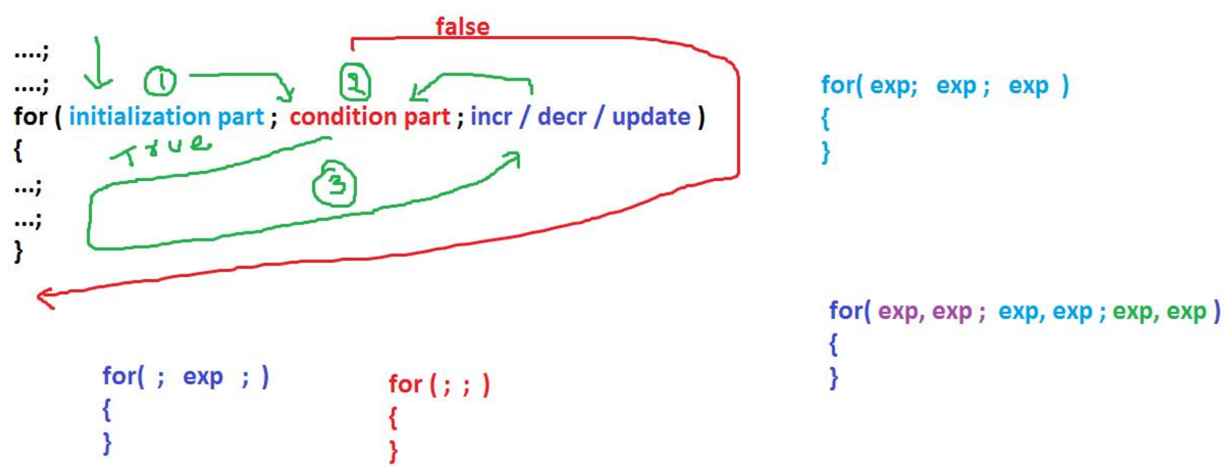
At first entry of for loop the initialization part is executed and later the test condition is checked. If the condition is true then the for block statements are executed. After completion of the block, the increment or decrement part is executed. Later once again the test condition is evaluated. If it is true then once again for block statements are executed. Like this the process is continued until the condition becomes false. Here the

initialization part is executed only once, at the time of loop beginning.

It is mandatory to maintain 2 semicolon (**;**) in a for loop.

If the for loop is having more than three expressions, it is mandatory to separate the expressions with **,** separator.

If the for loop is having less than three expressions, then leave the expressions with empty semicolon.



Printing given table:

For example 9^{th} table

9*1=9

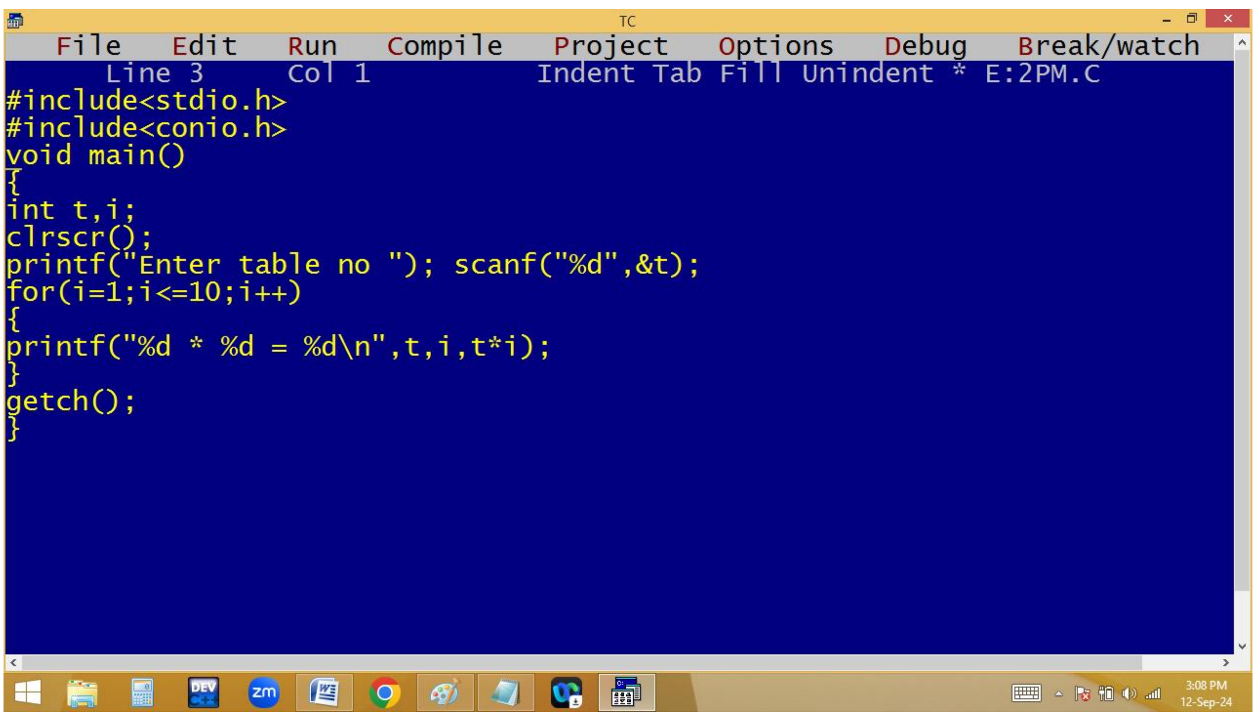
9*2=18

9*3=27

..

..

9*10=90



```
Enter table no 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```

```
Enter table no 100
100 * 1 = 100
100 * 2 = 200
100 * 3 = 300
100 * 4 = 400
100 * 5 = 500
100 * 6 = 600
100 * 7 = 700
100 * 8 = 800
100 * 9 = 900
100 * 10 = 1000
_
```

$$\underline{t} \times \underline{i} = \underline{t \times i}$$
$$9 \quad\;\; 1 \;\;=\;\; 9$$

```
for( i=1; i<=10; i++ )
{
    9 * 1 = 9
 p(" %d * %d = %d\n ", t, i, t * i );
}
```

Findig perfect no or not? Sum of factors is equal to given no

6 factors sum is 1+2+3 = 6

28 factors sum is 1+2+4+7+14=28

$6 \% 1 = 0$

$6 \% 2 = 0$

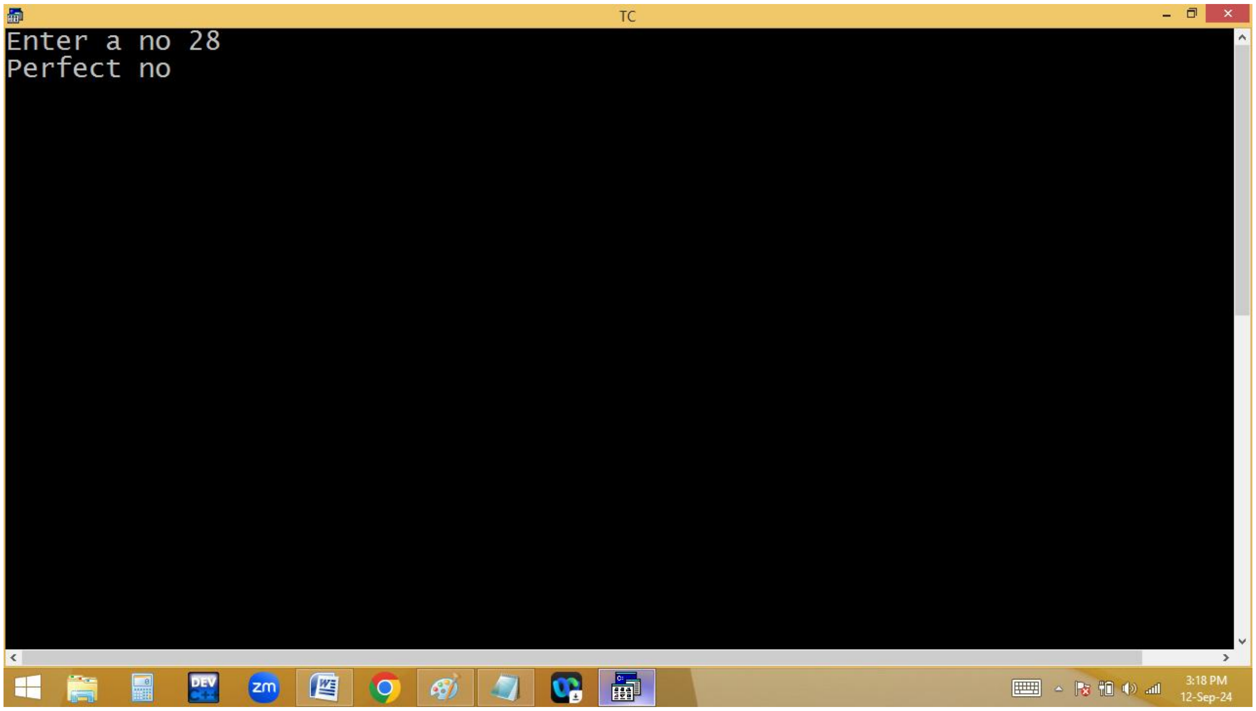
$6 \% 3 = 0$

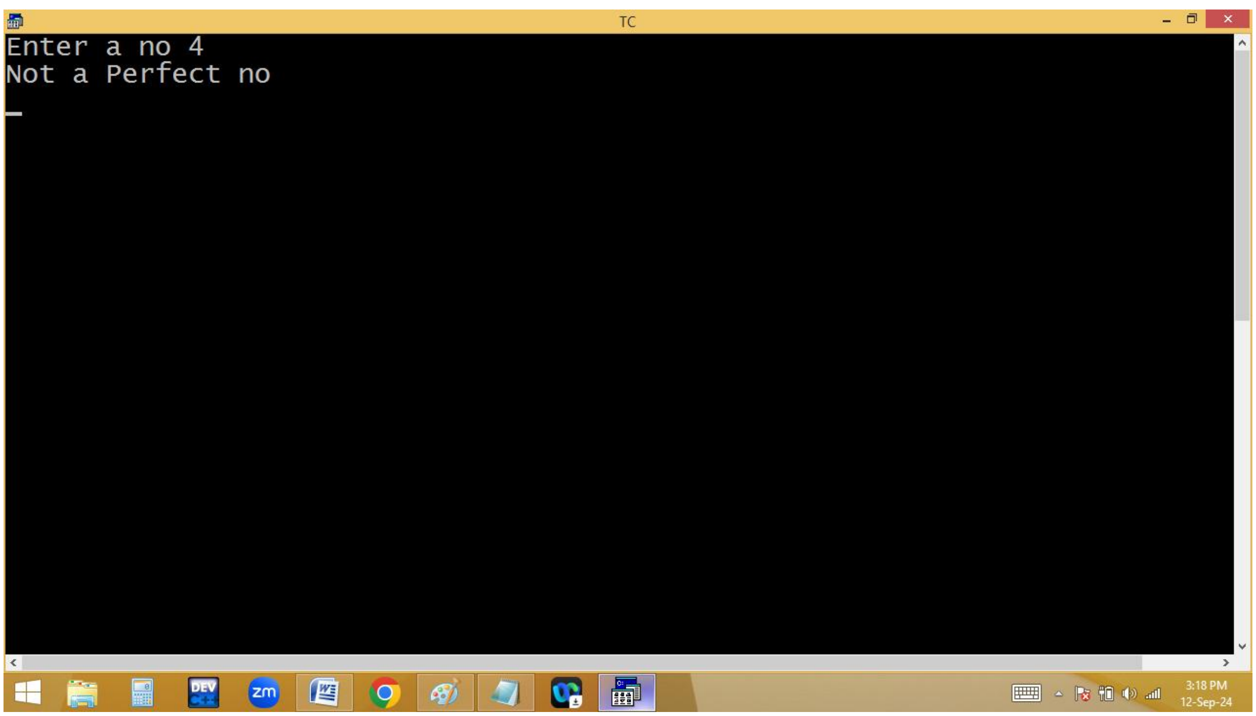
$\overline{6}$

4 factors sum is 1+2=3 🡨 not a perfect no

```
   File    Edit    Run    Compile    Project    Options    Debug    Break/watch
        Line 12     Col 44  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n/2;i++)
{
if(n%i==0)s=s+i;
}
puts(n==s?"Perfect no":"Not a Perfect no");
getch();
}
```

```
Enter a no 6
Perfect no
```


TC
Enter a no 28
Perfect no
3:18 PM
12-Sep-24


TC
Enter a no 4
Not a Perfect no
3:18 PM
12-Sep-24

```
for(i=1;i<=n/2;i++)
{
if(n%i==0) s=s+i;
}
puts(n==s?"Perfect":"Not");
```

n	i<=3	s

6 % 1 = 0

6 % 2 = 0

6 % 3 = 0

4 <= 3

0 + 1 + 2 + 3 = 6

4 % 1 = 0

4 % 2 = 0

0

1

2

3

Finding prime / composite no?

count of factors is 2 or the no divisible with 1 and itself only is called prime.

$2 \% 1 = 0$

$2 \% 2 = 0$

$3 \% 1 = 0$

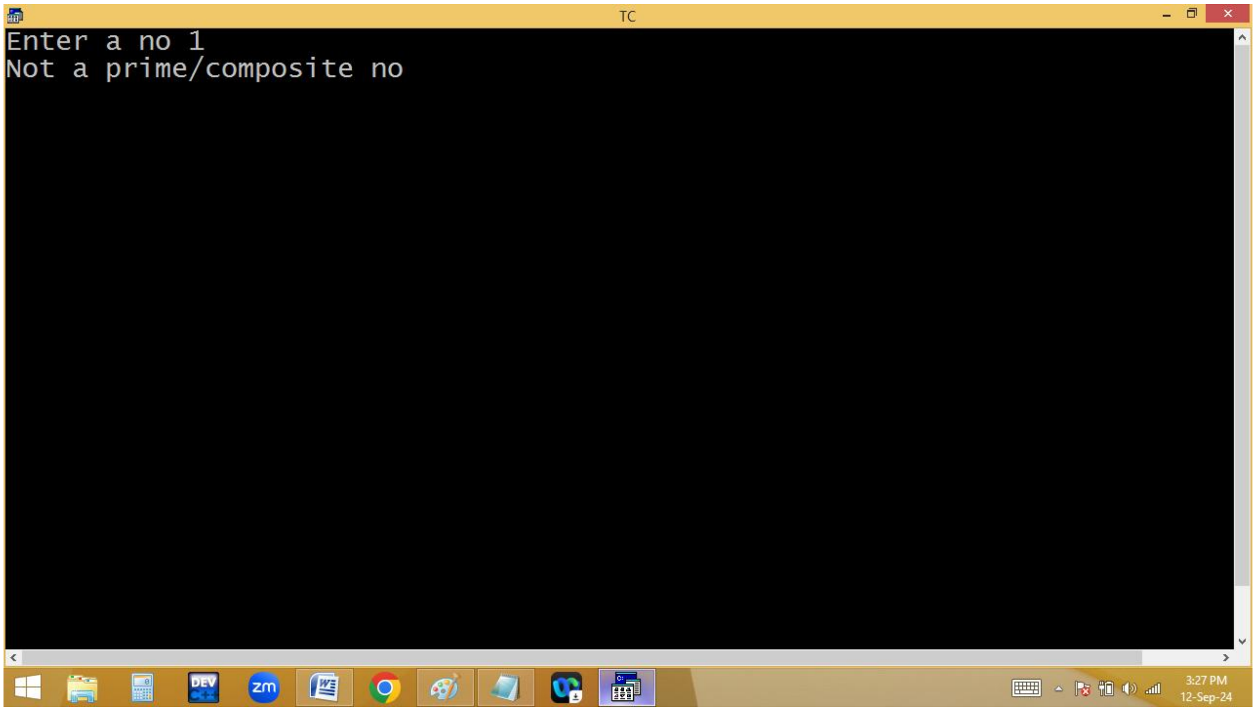
$3 \% 3 = 0$

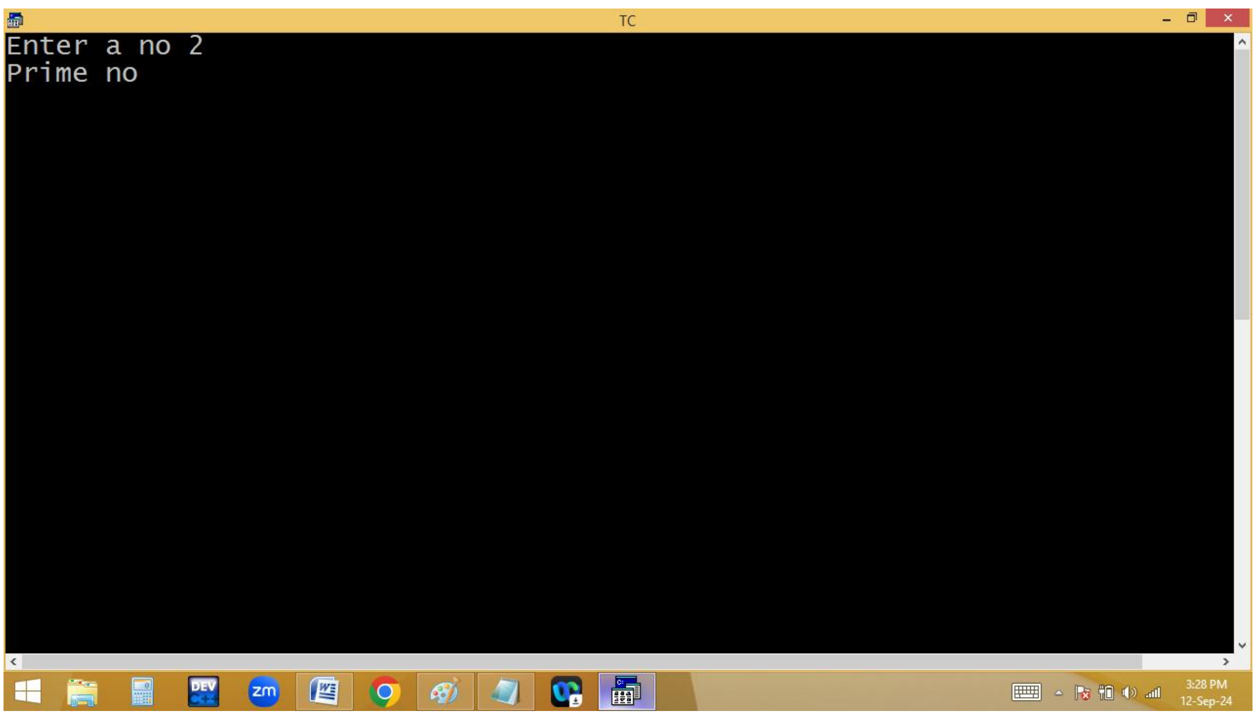
$4 \% 1 = 0$

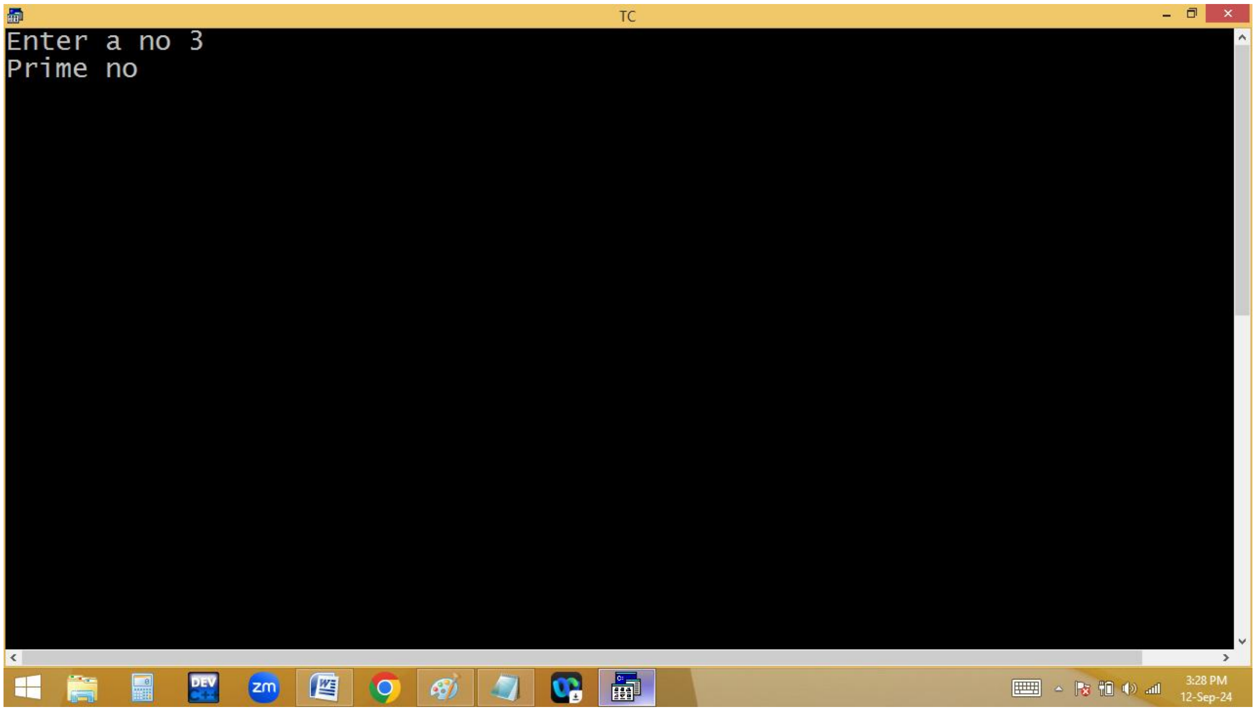
$4 \% 2 = 0$

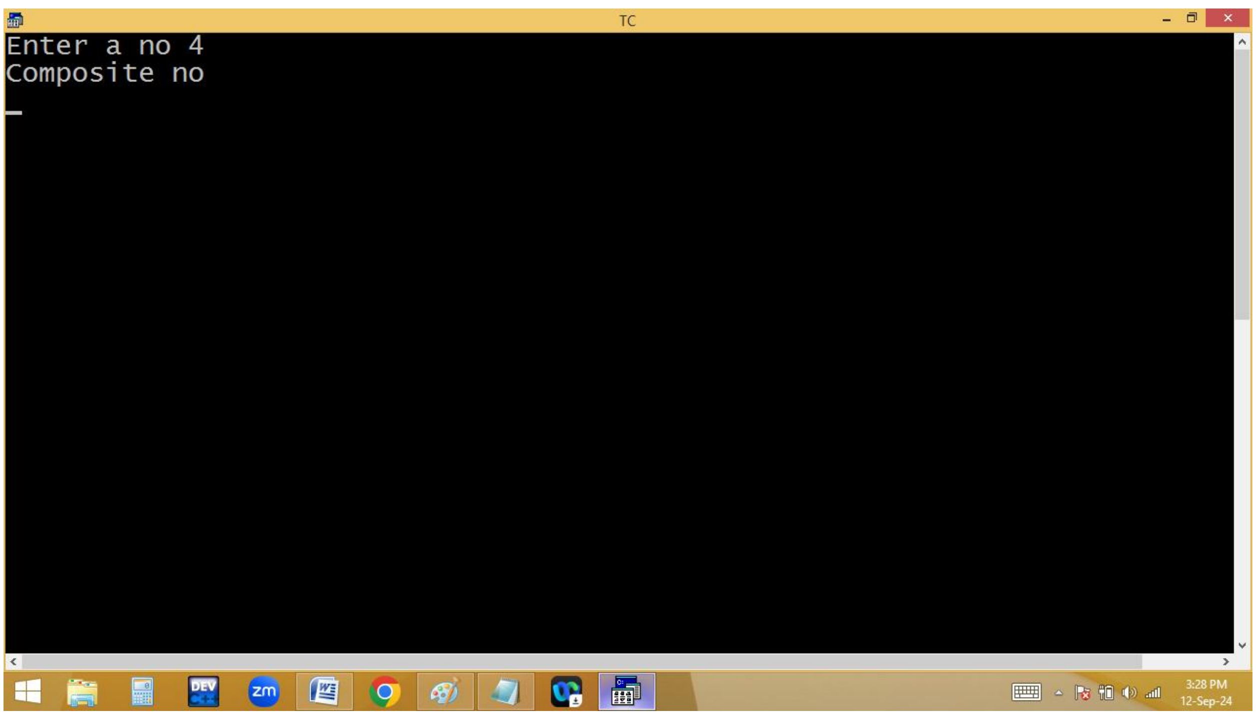
$4 \% 4 = 0$

```
TC
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 16    Col 2   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,c=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for(i=1;i<=n;i++)
{
if(n%i==0)c++;
}
puts(c==2?"Prime no":"Composite no");
}
getch();
}
```


TC
Enter a no 1
Not a prime/composite no
3:27 PM
12-Sep-24

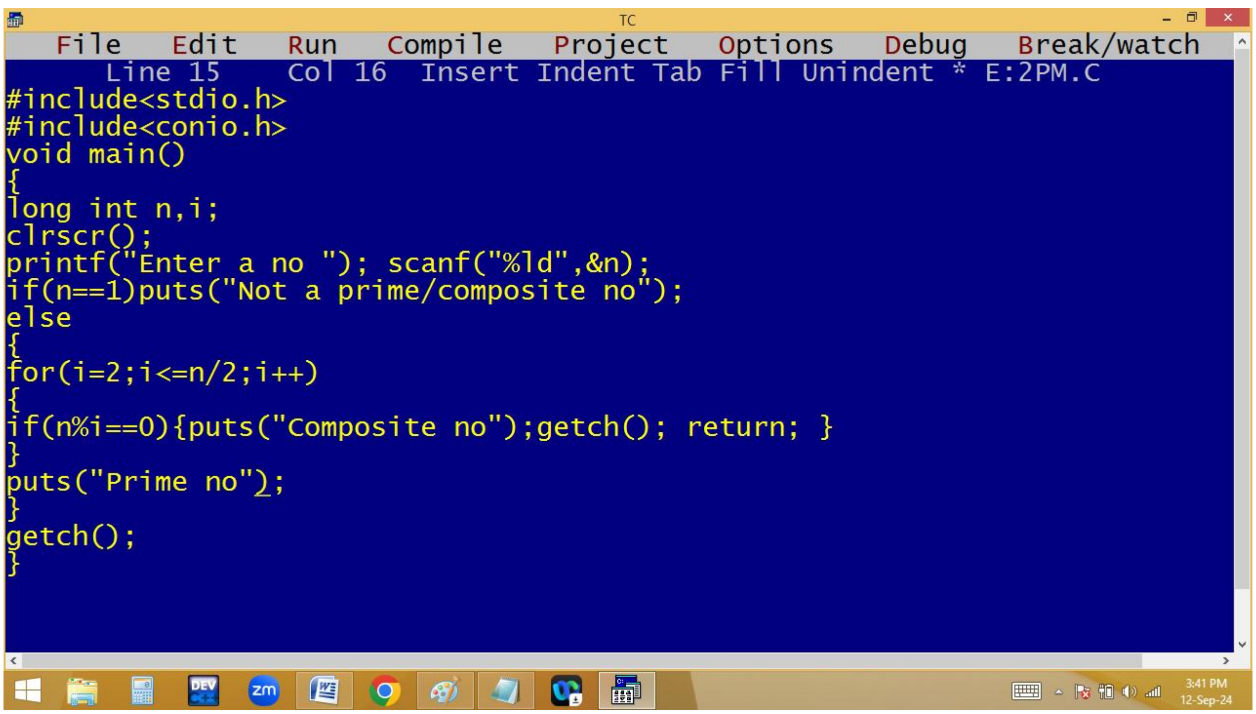

TC
Enter a no 2
Prime no
3:28 PM
12-Sep-24

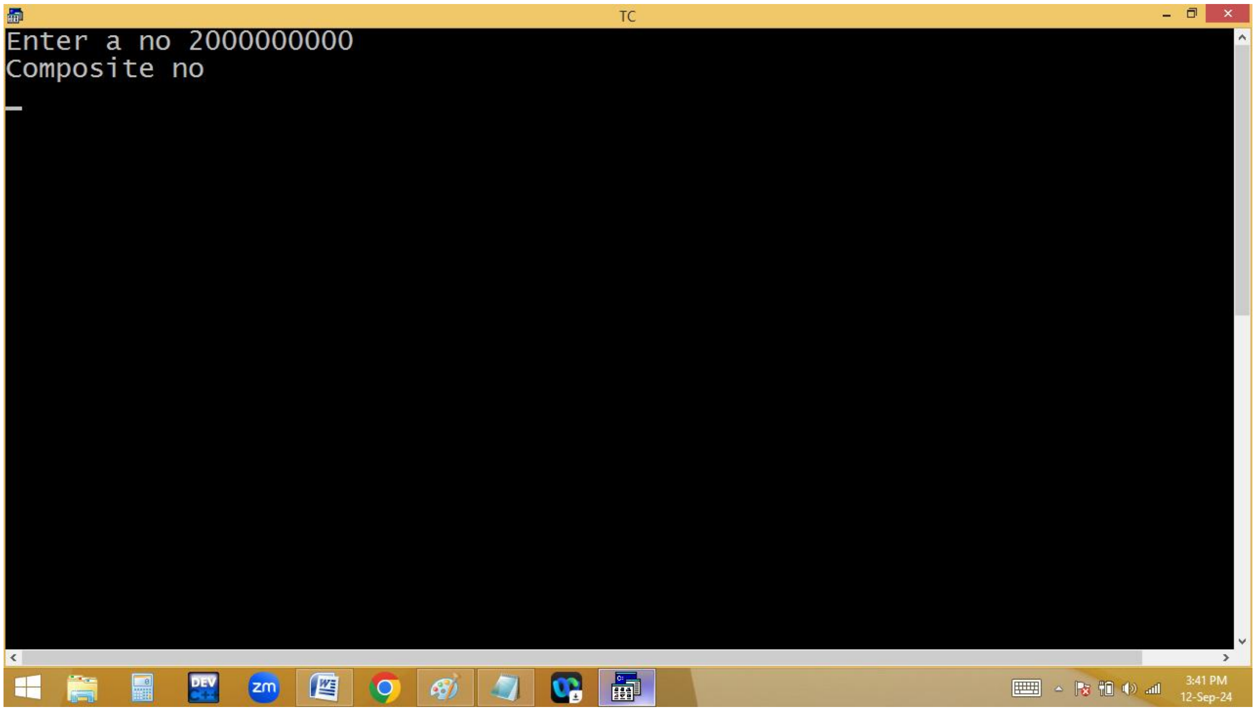

TC
Enter a no 3
Prime no
3:28 PM
12-Sep-24

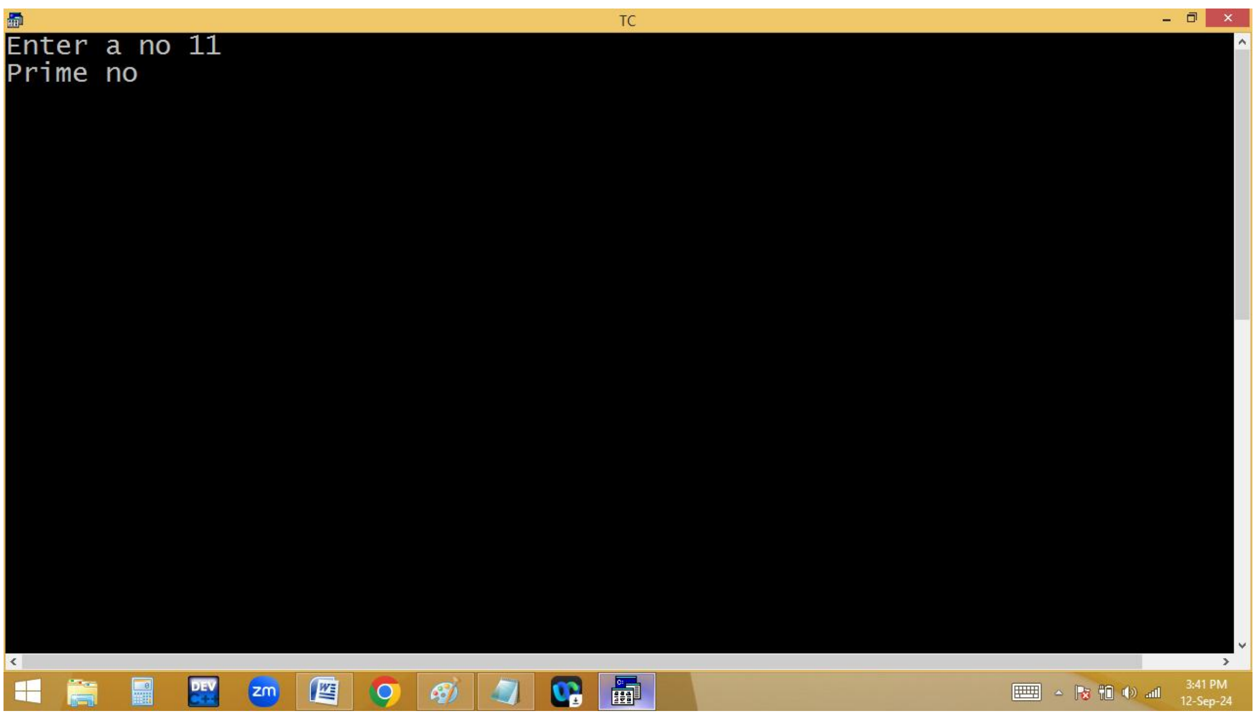

TC
Enter a no 4
Composite no
3:28 PM
12-Sep-24

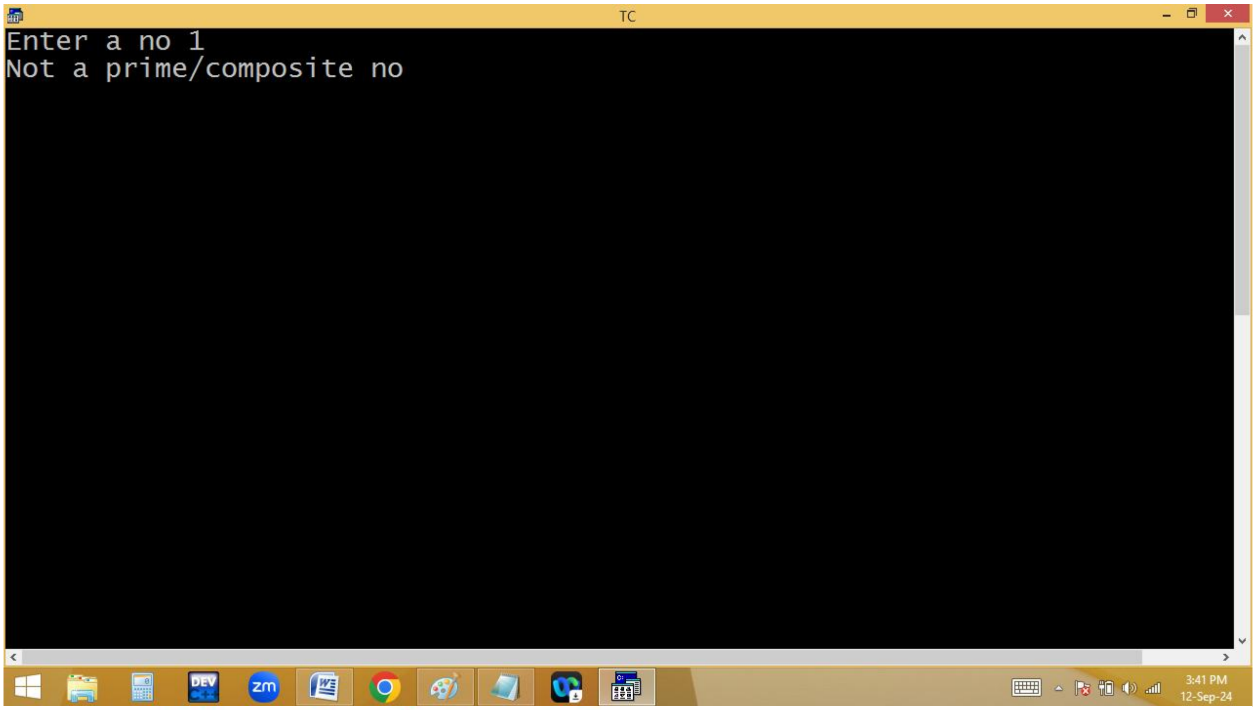
```
for( i=1; i<=n; i++ )
{
if(n%i==0)c++;
}
puts(c==2?"Prime":"composite");
```

n		i	c
3	%	1 = 0	0 1 2
3	%	2 = 1	
3	%	3 = 0	
4	%	1 = 0	1 2 3
4	%	2 = 0	
4	%	4 = 0	




TC
Enter a no 2000000000
Composite no
3:41 PM
12-Sep-24


TC
Enter a no 11
Prime no
3:41 PM
12-Sep-24



11 - ~~2~~ ~~3~~ ~~4~~ ~~5~~

10 - ~~1~~ 2 5 ~~10~~

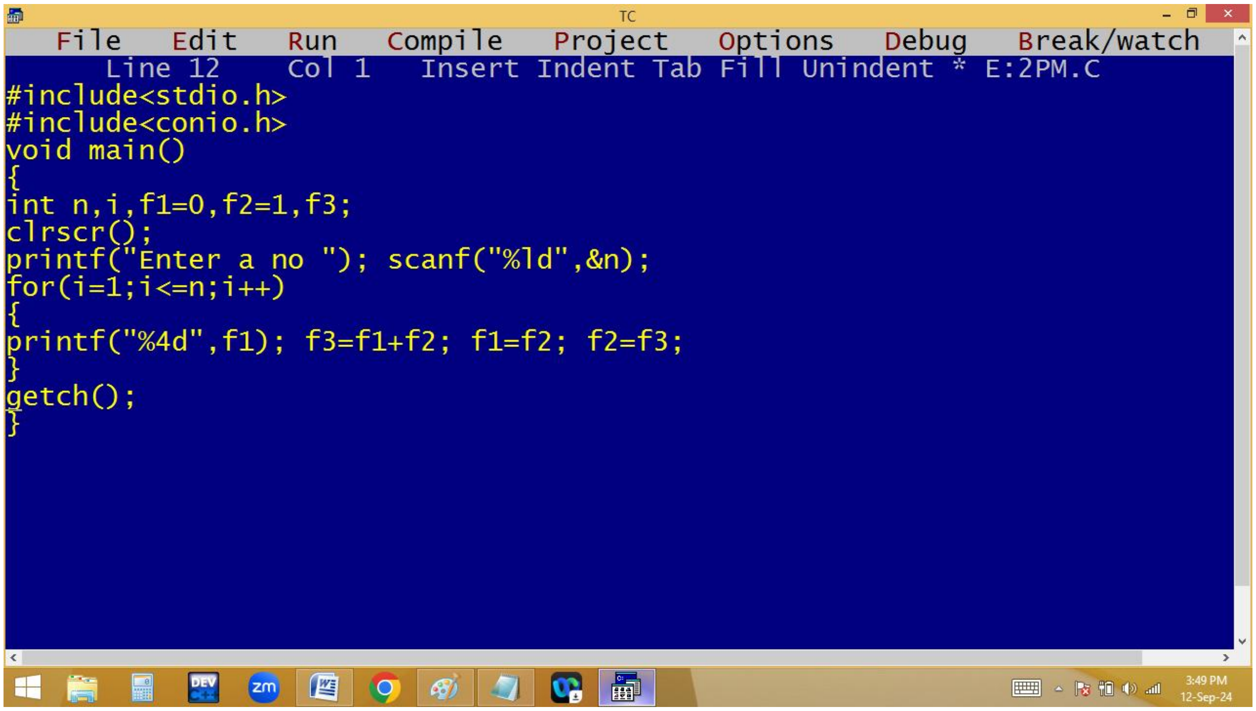
100 - ~~1~~ 2 4 5 10 20 25 50 ~~100~~

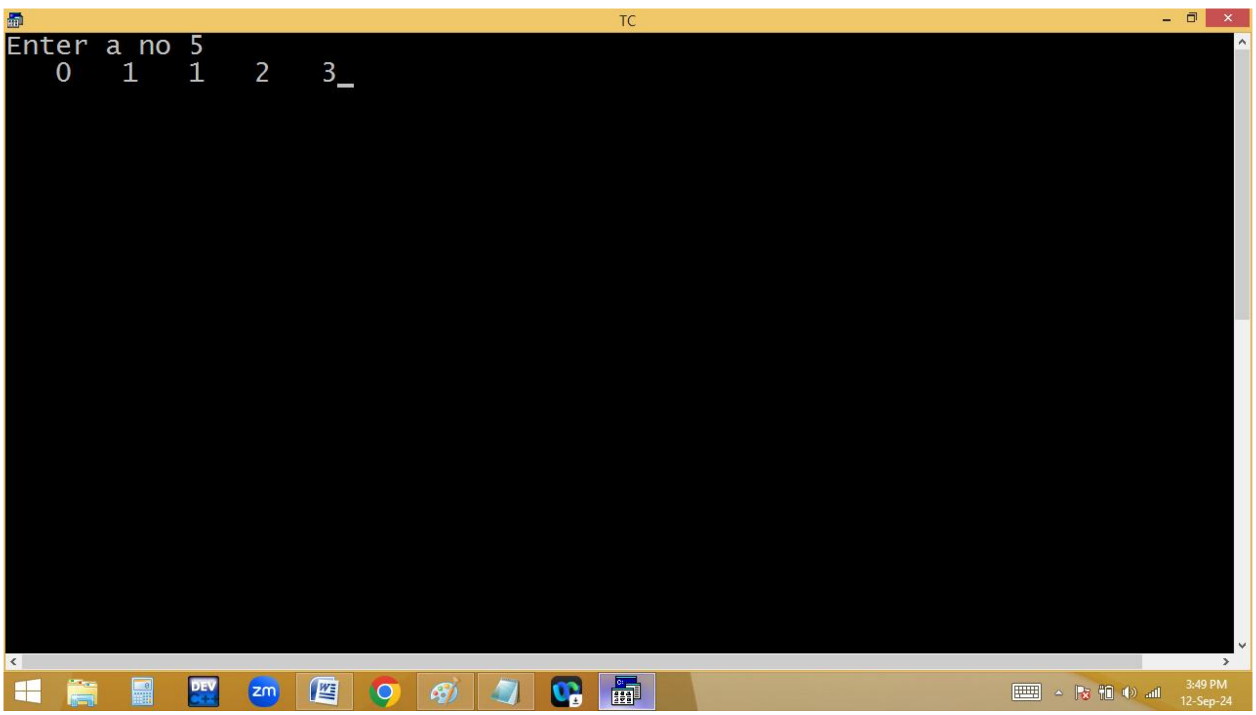
5 - ~~1~~ ~~5~~

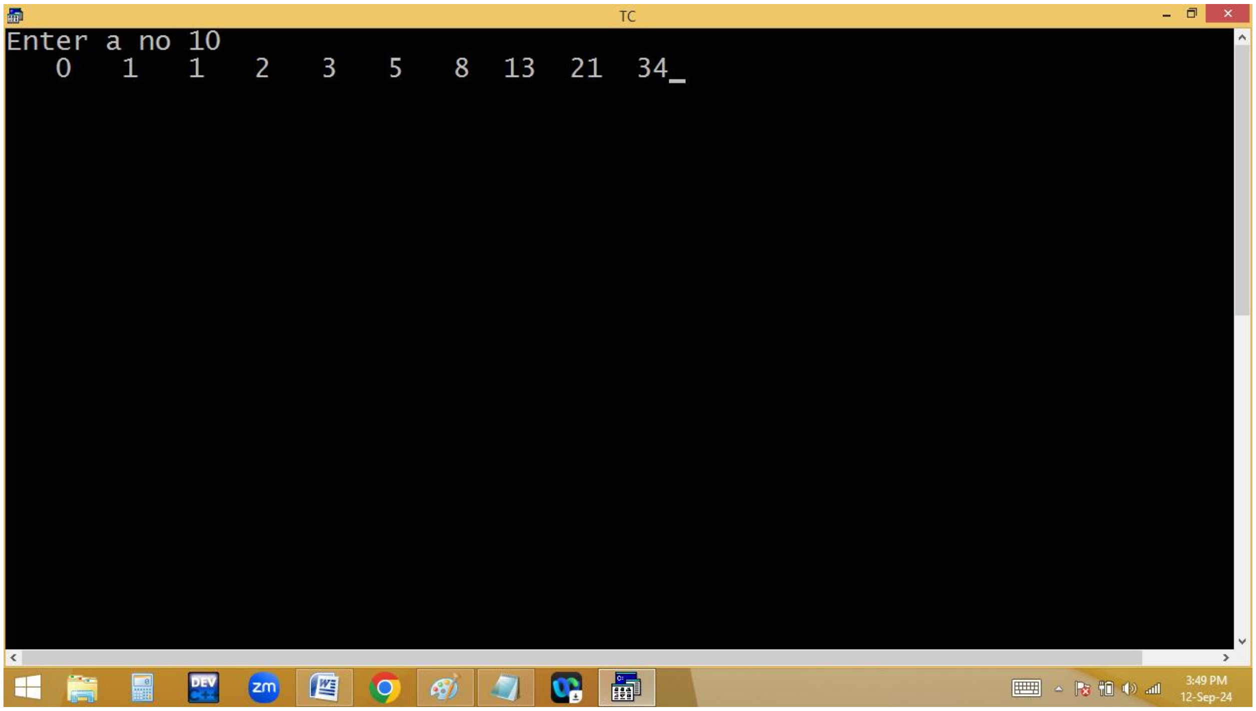
```
for( i=2; i<=n/2;i++)
{
if(n%i==0)p("com"); return;
}    p(prime);
```

Fibonacci series:

N = 5 → 0 1 1 2 3


TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,f1=0,f2=1,f3;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
for(i=1;i<=n;i++)
{
printf("%4d",f1); f3=f1+f2; f1=f2; f2=f3;
}
getch();
}
3:49 PM
12-Sep-24


TC
Enter a no 5
0 1 1 2 3_
3:49 PM
12-Sep-24


TC
Enter a no 10
0 1 1 2 3 5 8 13 21 34
3:49 PM
12-Sep-24